Lesson 3.1 — Introduction to Polynomial Functions

Big idea: A polynomial is a sum of terms ax^n with whole-number exponents. The degree, leading coefficient, and constant term carry most of the information about its shape and end behavior.

Quick review

Polynomial form: $P(x) = a_n x^n + a_{n-1} x^{n-1} + ... + a_1 x + a_0$, with non-negative integer exponents.
Degree: highest exponent. **Leading coefficient:** coefficient of the highest-degree term.
By degree: 0 = constant, 1 = linear, 2 = quadratic, 3 = cubic, 4 = quartic, 5 = quintic.
Not polynomials: negative exponents, fractional exponents, variables in the denominator, variables under a root.

Worked example

Worked example. Classify $P(x) = -5x^4 + 2x^3 - x + 7$.

Degree 4 (quartic), leading coefficient –5, constant term 7. It's a polynomial.

Warm-ups

1. State the degree and leading coefficient of $P(x) = 4x^3 - 2x^2 + x - 9$.
2. Is $f(x) = 1/x + 5$ a polynomial? Why or why not?
3. Is $g(x) = \sqrt{x} - 3$ a polynomial? Why or why not?

Practice

1. Classify $P(x) = 7x^5 - 3x^2 + x$ by degree and name (quartic, etc.).

2. Write a polynomial of degree 4 with leading coefficient –2 and constant term 5 (any other terms allowed).

3. Evaluate $P(x) = x^3 - 2x^2 + 4$ at $x = -2$.

Challenge

1. Decide whether each is a polynomial. If yes, give degree and leading coefficient. (a) $f(x) = 3x^2 - \sqrt{2} \cdot x + 1$ (b) $g(x) = 5x^{2/3} - x$ (c) $h(x) = (x^3 + x)/2$.

2. A polynomial P has degree 3 and $P(1) = 0$, $P(-1) = 0$, $P(2) = 0$, $P(0) = 6$. Find $P(x)$.

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Answer key — Lesson 3.1

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. Degree 3, leading coefficient 4.
2. No — $1/x$ has a negative exponent (x^{-1}) which is not allowed.
3. No — $\sqrt{x} = x^{1/2}$ has a fractional exponent.

Practice

1. Quintic (degree 5).
2. Many possible. Example: $-2x^4 + 3x^2 + 5$.
3. $(-2)^3 - 2(4) + 4 = -8 - 8 + 4 = -12$.

Challenge

1. (a) Polynomial. Degree 2, leading coeff 3 (the $\sqrt{2}$ is fine — it's a real-number coefficient). (b) NOT a polynomial (fractional exponent 2/3). (c) Polynomial. Degree 3, leading coeff 1/2.

2. Zeros at 1, –1, 2: $P(x) = a(x - 1)(x + 1)(x - 2)$. $P(0) = a(-1)(1)(-2) = 2a = 6 \Rightarrow a = 3$. So $P(x) = 3(x - 1)(x + 1)(x - 2)$.

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